Jugal Kakkat

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Skills

Languages	Python, C, C++, Bash
Frontend	HTML5, CSS3, JavaScript, React, Jinja2, Streamlit
Backend & APIs	Node.js, ExpressJS, FastAPI, REST APIs, JWT Authentication
Databases	MySQL, PostgreSQL, MongoDB
Tools & Systems	Git, GitHub, Linux, Unix Shell
Testing & Debugging	pytest, Streamlit AppTest, DeepEval
Agentic AI	LangGraph, LangChain, CrewAI
Observability	Langfuse
RTOS	Zephyr RTOS

Work Experience

Embedded Software Intern - Linumiz [Certificate] May 2025 - Jul 2025

- Implemented TCP and UDP communication between a Zephyr RTOS client and a Linux host
- Enabled transfer of llext modules to the LittleFS file system via custom shell commands
- Researched Zephyr's boot flow, initialization sequence, and memory management across multiple boards
- Gained hands-on experience with dynamic module loading and Zephyr subsystem configuration

Projects

Sherpa - Multi-Agent AI Study Companion Live | Code

- Built a **4-agent** learning pipeline in **LangGraph** (planner, explainer, quiz, coach) with **human-in-the-loop** approval and **SQLite** checkpointing for crash-safe session resume, running on a self-hosted **Qwen2.5-7B** via **Ollama** at **$0** API cost.
- Grounded explanations in the user's own notes through an agentic tool-calling **RAG** loop over **2 MCP servers** (**7** tools: path-traversal-safe file search and per-session memory), scoring **1.0** faithfulness in **DeepEval** LLM-as-judge evals.
- Eliminated hallucinated quiz answers by running generated code in a **2-layer sandbox** (AST allowlist + resource-limited subprocess with audit-hook tripwire); correct answers once graded **0%** now score **85%**, hardened against **25+** escape tests.
- Deployed on **Streamlit Cloud** with an on-demand **Kaggle T4 GPU** behind an authenticated **ngrok** tunnel: one-click start in **∼2 min**, auto-shutdown after **30 min** idle; backed by **424** unit tests and **14** live LLM evals.

Pathfinder - Autonomous Web Research Agent Live | Code

- Built a **ReAct** research agent in **LangGraph** on **gpt-oss-20b** (**Groq**) that plans queries, searches the web via **DuckDuckGo** and writes a sourced summary.
- Capped LLM cost with a configurable search budget (**1**–**8**) held in graph state, forcing the agent to summarise once it is spent.
- Streamed every graph step live to a **Streamlit** UI using LangGraph's update and custom stream modes, including search retries with **10/30/60 s** backoff against rate limits.
- Deployed on **Streamlit Community Cloud** with browser-persisted history and Markdown export.

Shrinx - URL Shortener Live | Code

- Built a full-stack URL shortener with **FastAPI**, **MongoDB Atlas** and **Jinja2** server-side rendering.
- Engineered a collision-safe short-code generator using Python's `secrets` module over a **62**-character alphabet, yielding a **3.5 trillion**-code keyspace.
- Implemented stateless **JWT** authentication with **bcrypt** password hashing, `HttpOnly` + `SameSite` session cookies and **7**-day token expiry, offloading CPU-bound hashing to a thread pool to keep the async event loop non-blocking under concurrent requests.

Education

2023 – Present B.Tech at **Indian Institute of Information Technology Kottayam**
2022 Class 12 CBSE at **Kailasanadha Vidyanikethan Mannuthy**

LEADERSHIP EXPERIERNCE

Technical Club Mentor - IIIT Kottayam Sep 2025 - Aug 2026

- Conducted the Udbhav All-India Inter-IIT Hackathon round 1 for 100+ participants at IIITK
- Organised Techashy 2026, a national-level hackathon attracting participants across institutions
- Led workshops, CTFs, and internal hackathons to build technical culture in the college community

Certifications

HackerRank SQL (Intermediate) [Certificate] HackerRank Python (Basic) [Certificate]